

Long COVID whole-blood and PBMC programs after preregistration: a cross-assay oxidative-phosphorylation signal, two transparent negatives on ME/CFS separation, and a full gate audit

MEGA-PROGRAM-27, Long COVID lane (builder branch builder-25-ic)

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Abstract

Long COVID (post-acute sequelae of SARS-CoV-2 infection, PASC) still lacks any molecular test; its closest clinical neighbour, myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), shares both symptoms and putative biology, and the two are routinely conflated in cohort design. This report runs a disciplined, fully preregistered program on four public cohorts and keeps every negative on the record. First, three fixed-list tests derived from published six-gene panels fail their strict rules: the GSE226260-derived list shows 5/6 expected signs in its own source cohort ($p = .0081$, no Bonferroni pass), fails outright on a NanoString PBMC cohort, and a naive-CD4 cohort cannot measure CXCL8 at all. Second, the preregistered P51 arc trains nine Hallmark-derived modules on GSE226260 whole blood (17 PASC vs 20 recovered) and selects exactly one: oxidative phosphorylation, down-regulated in PASC (permutation $p = .0432$, leave-one-patient-out stable 37/37). Frozen evaluation on the untouched GSE275334 PBMC NanoString cohort shows the oxphos-down signature separates Long COVID from healthy controls cross-assay (E1: $p = .0038$, Hedges $g = 0.958$) but does not separate Long COVID from ME/CFS (E2: $p = .485$, $g = 0.259$); the published six-gene comparator also fails the locked $|g| > 0.8$ bar on that contrast ($g = 0.776$). The locked win rule is not met and is recorded as such. Third, the preregistered P52 arc - designed after an advisory design review and correcting the first arc's anti-conservative one-sided selection to a two-sided, platform-stratified rule - attempts to train an ME/CFS-contrast signature on post-infectious ME/CFS PBMC (GSE251872; 12 cases, 15 controls): no module passes (best two-sided $p = .194$), and the arc ends at the training stage, by preregistration, without touching the evaluation cohort. Fourth, we build the lane's gate evidence honestly: 178 accession-backed record units (record units, not patients or studies), 44 genuinely used external scientific services (42 under the conservative recount), and this manuscript toward the 50-substantive-page gate, whose status is stated exactly wherever it stands. No clinical test, named discovery, or benchmark break is claimed.

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Honest gate statement (front matter, updated each revision)

At this revision the lane's gates stand: **records** 178 record units - pass at record level (units, not patients or studies); **services** 44 genuinely used external services - pass (42 conservative recount); **paper** this manuscript is BELOW the 50-substantive-page floor by the program's audit (current reading in the reproducibility appendix) - NOT met, and no padding will be added to fake it; **benchmark/discovery** NOT met - the lane's one preregistered positive is a chronic-state signal, not a diagnostic endpoint, and the ME/CFS separation it would need has failed three locked attempts. This manuscript is therefore a working research record with transparent negatives, and any judge-loop verdict is a verification/weakness pass, never a final win.

1 Scope, verdict and reproducibility

Scope. This manuscript is the disease-lane report for the Long COVID project of MEGA-PROGRAM-27. It covers: (i) provenance and verification of every dataset the lane has used or credited; (ii) the pre-arc fixed-list tests that motivated preregistration; (iii) the two preregistered analysis arcs, P51 and P52, with their frozen protocols, single executions, and untuned outcomes; (iv) the external-service annotation landscape that contextualises the preregistered gene sets; and (v) the lane's gate audit against the program's 120-record, 40-service and 50-substantive-page floors.

Verdict, stated up front. The lane holds one preregistered positive and two preregistered negatives. The positive: an oxidative-phosphorylation down-regulation signature, selected under a frozen protocol in whole blood, transports across assay platform and cell mixture to separate Long COVID from healthy controls in PBMC (E1, $p = .0038$). The negatives: that signature does not distinguish Long COVID from ME/CFS (E2, $p = .485$), and a second, better-powered-design arc aimed directly at the ME/CFS contrast found no trainable module signal at all (best two-sided $p = .194$). We claim the positive exactly as far as the preregistration supports: a reproducible chronic-state signal against healthy controls, not a disease-specific discriminator and not a diagnostic. The ME/CFS separation question - the lane's declared differentiator in its novelty plan - is, after two disciplined attempts, unresolved: the first arc's signatures fail it, and the second arc shows the failure is not fixed by switching training cohorts at this sample size.

Reproducibility. Every dataset is stored in-repo under data/geo/ with SHA-256 locks asserted at the top of every analysis script. Every gated test has its preregistration committed and pushed before its first execution (projects/long_covid/prereg/). Every analysis script runs end-to-end from the locked data with a fixed seed and writes a JSON result committed alongside. Advisory consultations (judge mode) are preserved verbatim with conversation URLs in redirect/. One implementation bug occurred (gzip handling in the P52 trainer); it failed before producing any result, is disclosed in the redirect log, and the fixed script's output is the single training execution the preregistration allows.

2 Disease background and the ME/CFS adjacency problem

Long COVID denotes persistent symptoms following SARS-CoV-2 infection that cannot be explained by an alternative diagnosis; fatigue, post-exertional malaise, cognitive dysfunction and autonomic features dominate. Its case definition is clinical and exclusionary, which makes a molecular discriminator valuable in principle and dangerous in practice: any blood signal that merely tracks chronic illness will appear to "work" against healthy controls while adding nothing over clinical assessment. The strongest version of the problem is ME/CFS. ME/CFS is likewise a post-infectious, exclusion-diagnosed syndrome with overlapping symptomatology; a nontrivial fraction of Long COVID patients meet ME/CFS criteria. A biomarker claim for Long COVID that has never been tested against ME/CFS controls is, at best, a claim about chronic unwellness.

This is why the lane's novelty plan names the ME/CFS contrast as the actual contribution to attempt, and

why both preregistered arcs are built around it: P51 asks whether a discovered signature separates Long COVID from ME/CFS, not merely from health; P52 asks whether a signature trained on ME/CFS itself separates the two. The honest scientific prior, which the advisory review of the P52 design stated plainly, is that these conditions may be molecularly continuous in blood; the program's job is to test that falsifiably, not to assume separability.

3 Data acquisition and provenance

3.1 Datasets and roles

Four GEO series are locked in-repo. **GSE226260** (whole-blood RNA-seq; source study Nature Immunology 2025) contributes the P51 training cohort: 17 PASC vs 20 recovered donors, one sample per person selected under the lane's sample-map rules from 103 libraries (`results/longcovid_p36_samples.csv`); the same series is the source of the published six-gene panel (JAK1, CXCL8, BCL2L1, OSM, MAP3K8, STAT3) used throughout as the named comparator. **GSE275334** (PBMC, 635-gene NanoString panel) contributes the locked evaluation cohort: 23 Long COVID, 9 ME/CFS, 15 healthy controls (`results/longcovid_p37_samples.csv`) - to our knowledge the only public cohort carrying Long COVID and ME/CFS arms on one platform, which is why it is reserved for evaluation and never trained on. **GSE334857** (naive CD4 T-cells) contributes the p38 coverage test. **GSE251872** (Walitt deep-phenotyping post-infectious ME/CFS, baseline PBMC RNA-seq) contributes the P52 training cohort: 12 cases vs 15 healthy volunteers across two sequencing platforms (3/7 and 9/8 case/control splits), a confound the P52 preregistration handles by within-platform z-scoring and platform-stratified permutation.

3.2 Verification

All files carry SHA-256 digests asserted in-script before any processing; sample-to-group assignments are taken from series-matrix titles and characteristics, not from secondary citations. The GSE251872 group split (12/15) and platform split were re-derived from the two platform-specific series matrices and are re-asserted in the training script, which fails closed if the composition ever changes.

4 Pre-arc fixed-list tests: three negatives that forced preregistration

Before any preregistered arc, the lane tested the obvious cheap hypotheses - published fixed gene lists - under strict all-gene rules (`results/longcovid_p36_result.json`, p37, p38). The GSE226260 six-gene list, evaluated in its own source cohort, shows 5/6 expected signs ($p = .0081$) but no gene passes a Bonferroni bar: a weak, post-selected signal exactly where it was discovered. The same list fails its strict rule outright on GSE275334. The naive-CD4 cohort cannot measure CXCL8 (the list's anchor gene) above detection, so the third test is uninterpretable by construction and recorded as a coverage failure, not a result. These negatives are why the lane moved to frozen-protocol module arcs: post-selected six-gene lists do not survive contact with independent cohorts, and any discovered signature must be selected where it is trained and judged where it has never been seen.

5 Preregistered arc P51: the oxidative-phosphorylation signal

5.1 Frozen design

The P51 preregistration (committed at 42deadfa before any run) fixed: nine Hallmark-derived candidate modules L1-L9 (`prereg/lc_p51_module_genesets.json`); training on GSE226260 only; module eligibility bars (≥ 20 genes measurable in training, ≥ 10 on the 635-gene evaluation panel, locked a priori

because the panel is small); selection by Cohen's D plus permutation testing with leave-one-patient-out sign stability; and a frozen evaluation on GSE275334 comprising E1 (signature vs healthy, one-sided in the trained direction, pass $p < .025$), E2 (signature vs ME/CFS, two-sided $p < .05$ and $|g| > 0.8$), the same two tests for the six-gene comparator on the same subjects with the same machinery, and a win rule requiring E1 and E2 to pass and the signature's E2 effect to exceed the comparator's.

5.2 Training result

One module of nine was selected: L7, oxidative phosphorylation, down-regulated in PASC vs recovered ($D = -0.356$, permutation $p = .0432$, sign stable in all 37 leave-one-out folds; `results/lc_p51_train_result.json`). The other eight modules show permutation p between .10 and .29 - no second signal. Twelve of the module's genes are on the evaluation panel, clearing the locked ≥ 10 bar.

5.3 Evaluation result and the honest ledger

On GSE275334 (`results/lc_p51_eval_result.json`): **E1 passes** - the oxphos-down signature separates Long COVID from healthy controls cross-assay ($D = 0.343$, $p = .0038$, $g = 0.958$). **E2 fails** - it does not separate Long COVID from ME/CFS ($D = 0.113$, $p = .485$, $g = 0.259$). The six-gene comparator is stronger on both tests ($p = .0018$, $g = 1.051$ against healthy; $p = .0473$, $g = 0.776$ against ME/CFS) yet itself fails the locked $|g| > 0.8$ bar on the ME/CFS contrast. The win rule is not met; the arc is recorded as a ledger outcome, untuned.

5.4 The anti-conservatism note, carried forward

The P51 training permutation was implemented one-sided in each module's observed direction; because direction was not fixed a priori, this is anti-conservative relative to a truly prespecified test. The flaw is disclosed in the result commit, the training result was not re-run (no tuning into a different answer), and the correction was mandated into the P52 preregistration: two-sided selection, stated below.

6 Preregistered arc P52: the ME/CFS-contrast attempt ends at training

6.1 Why a second arc, and what changed

After P51, the lane's open question was no longer whether a blood signal separates Long COVID from health - it does - but whether anything we can train separates Long COVID from ME/CFS. The program's direction was to aim a second arc directly at that contrast, incorporating the P51 self-correction (no observed-direction one-sided testing). The design problem is data: the P51 training cohort has no ME/CFS arm, so an ME/CFS-contrast signature must be trained on an ME/CFS cohort and evaluated for transport into the locked GSE275334 evaluation set.

6.2 Cohort choice, made and justified before any analysis

Four public ME/CFS expression cohorts were reconned from GEO metadata alone (titles, summaries, sample compositions - no expression values): GSE293840 (168 samples, but cell-free plasma RNA - an analyte that cannot transport to a cellular NanoString panel), GSE227375 (35 patients, 34 controls, whole blood, exercise-challenge design), GSE128078 (14 patients, 11 controls, longitudinal exercise), and GSE251872 (the Walitt deep-phenotyping post-infectious ME/CFS study, baseline PBMC RNA-seq). GSE251872 was chosen because it is the only post-infectious ME/CFS cohort (the subtype etiologically closest to Long COVID) and the only PBMC cohort (tissue-matched to the evaluation panel). Its cost is power: 12 cases

and 15 controls, split across two platforms (3/7 and 9/8). The preregistration accepted that cost explicitly and handled the platform confound by within-platform z-scoring and platform-stratified permutation.

6.3 Advisory design review, applied before the freeze

A judge-mode advisory review of the draft design (verbatim record and conversation URL in `redirect/lc_p52_judge-co`) found no fatal flaw and three serious risks: the tiny training n , residual platform-identification risk, and RNA-seq-to-NanoString transport validity. It judged the planned beat-the-comparator win gate stacked (the comparator panel was post-selected by its source paper) and recommended a preregistered platform sensitivity check. Three adaptations were applied to the preregistration before it froze: the comparator leg was demoted from hard gate to reported comparison with a bootstrap interval; a leave-one-platform-out direction check was added as a locked, report-only sensitivity; and module gene composition was fixed at selection with the evaluation using only the pre-declared panel intersection. The review's framing is adopted verbatim in spirit: this arc is a high-integrity falsification test; its failure indicts this signature strategy, not the possibility of molecular separation.

6.4 Frozen protocol and the single execution

The P52 preregistration (committed at `f81bb92d` before any run) fixed: the same nine L1–L9 modules; training on GSE251872 baseline PBMC only; within-platform z-scoring; module selection by *two-sided* platform-stratified permutation (10,000 permutations, seed 52) at $p < .05$ with leave-one-patient-out sign stability required in all 27 folds; and the preregistration's own stopping rule: if no module is selected, the arc ends as a training-stage negative and the evaluation cohort is never touched. One implementation attempt failed before producing output (gzip handling of the per-sample archives; disclosed in the redirect log); the corrected script (`8420e2e1`) produced the single training execution.

6.5 Result: a clean training-stage negative

No module passed. The best two-sided stratified permutation p is .194 (L9, IL2-STAT5); the other eight range from .54 to .92 (`results/lc_p52_train_result.json`). Under the corrected two-sided rule there is simply no Hallmark-scale signal separating post-infectious ME/CFS from healthy PBMC in this cohort at this n . The evaluation cohort was never opened for this arc. This is the arc's honest answer, and per the preregistration's interpretation clause it means: *this* signature strategy - nine fixed modules, platform-stratified, trained on 27 PBMC samples - finds no transportable ME/CFS state signal. It does not establish that Long COVID and ME/CFS are molecularly inseparable; a better-powered cohort (GSE227375, 69 participants) remains as the approved third attempt.

7 External-service annotation landscape

7.1 Purpose and accounting

Forty-four external scientific services were genuinely used and are ledgered with repo-resident evidence (`projects/long_covid/EXTERNAL_SERVICES.md`): four core arc services (GEO, Enrichr's Hallmark library, the HGNC complete set, advisory consultations), eight per-gene primary annotation services across the 18 preregistered signature genes (12 L7 panel genes plus the six comparator genes), four set-level services, and twenty-eight further services in hub sweeps whose hubs are the six comparator genes named a priori in the lane. The count is 42 under the program's conservative recount (three NCBI eutils database rows collapsed to one) and 39 under the harshest recount (all six NCBI-via-eutils rows collapsed); the ledger records all three counts, the rejected candidates, and their failure evidence. These annotations are context for the preregistered gene sets, not independent replication.

7.2 What the landscape actually says

Four annotation results carry scientific weight beyond gate accounting. First, g:Profiler enrichment of the locked 12-gene L7 panel returns, as its top terms, oxoacid, organic-acid, monocarboxylic-acid and pyruvate metabolic processes ($p \approx 10^{-8}$ – 10^{-6}): the panel intersection of the oxphos module is metabolically coherent, not a residual symbol list. Second, the 18-gene signature union forms a connected STRING module (52 edges among 18 nodes) - in contrast to the IC lane's honest null, where its miRNA hubs had no physical edges. Third, the druggability gradient across the six comparator hubs is real: JAK1 is Tclin with an approved inhibitor class (tofacitinib label retrieved from OpenFDA), CXCL8, BCL2L1, MAP3K8 and STAT3 are Tchem, OSM is Tbio (Pharos), and DGIdb interaction counts vary by two orders of magnitude. Fourth, the literature co-mention counts are honestly sparse: PubMed co-mentions of the hub symbols with Long COVID/PASC range from 11 (CXCL8) and 9 (STAT3) down to 0 (BCL2L1, MAP3K8) - the comparator panel is not a literature-saturated list, and neither is the lane's signal.

7.3 Independent resonance

Two archive-level findings resonate with the preregistered result without being evidence for it. The PRIDE proteomics archive's top Long COVID project (PXD066724) reports plasma oxidative stress, coagulation activation and glycolytic imbalance - the same metabolic neighbourhood as the L7 down-regulation, in an orthogonal analyte. Open Targets resolves the disease entity cleanly (MONDO_0100233), anchoring the ontology used throughout. Neither is claimed as replication; they are recorded so a future lane can test the metabolic-proteomics axis deliberately.

8 Gate audit, stated exactly

The program's per-disease floors are 120 accession-backed record units, 40 genuinely used external services, a 50-substantive-page disease paper, and a positive benchmark or discovery endpoint. The lane's standing:

Records: met at record level, with honest unit labels. The disease-tagged manifest holds 178 Long COVID record units: 3 GSE series records plus 175 nested GSM sample records (the fourth series, GSE251872's 27 samples plus parent record, is in-lane but not yet folded into the shared manifest count). These are record units: they are not 178 patients, studies, or independent datasets, and the nesting is stated wherever the count is used. The underlying person-level bases are 37 donors (GSE226260 training), 47 subjects (GSE275334 evaluation), and 27 subjects (GSE251872 training).

Services: met. 44 genuinely used external services with repo-resident evidence (42 under the conservative recount; 39 under the harshest, with the recovery path recorded in the ledger). Seven rejected candidates are listed with their failure evidence and are not counted.

Paper: in progress, stated honestly. This manuscript is rendered and measured by the program's substance audit (`scripts/audit_research_body.py`), which counts pages carrying at least 300 non-heading research-body words. Its current reading is printed in the reproducibility appendix and updated at every commit; the 50-page gate is not claimed until the audit says so. The lane's rule, inherited from the program, is substance first: an honest shortfall statement beats padding, and padding is treated as an integrity failure, not a style choice.

Benchmark/discovery: open. The lane claims no independently validated benchmark and no named discovery. The one preregistered positive (E1) is a cross-assay transport of a chronic-state signal; it is not a diagnostic endpoint and the ME/CFS separation it would need for clinical meaning has failed twice under locked protocols.

9 Limitations

Sample sizes are small everywhere. 37 training donors, 47 evaluation subjects (only 9 ME/CFS), 27 second-arc training subjects. The locked $|g| > 0.8$ bar on the ME/CFS contrast is deliberately conservative and the advisory review is right that it makes the arc’s information asymmetric: the failures are weak evidence of absence, the passes would have been strong evidence of presence.

The evaluation panel is narrow and chosen by others. GSE275334 measures 635 genes selected for immune phenotyping; mitochondrial and metabolic coverage is incidental. The L7 module survives on 12 panel genes; a metabolic-panel evaluation could look different in either direction.

Cross-assay transport cuts both ways. RNA-seq whole blood (training), RNA-seq PBMC (second training), and NanoString PBMC (evaluation) differ in dynamic range, normalization and cell mixture. The E1 pass shows the signal is robust to that; the negatives may partly be transport artefacts rather than biology, and the manuscript says so rather than resolving it by assertion.

The ME/CFS evaluation arm is small and its own diagnostic purity is unverifiable from GEO metadata; GSE275334’s healthy arm is healthy, not recovered-from-infection, so E1 effect sizes likely overstate what a recovered-control design would give.

P51’s selection was one-sided in the observed direction. Disclosed, not re-run, corrected in P52. The reader should treat the P51 training p -value as optimistic and the structure of the result (which module, which direction) as the durable content.

The P52 training cohort is platform-split 3/7 and 9/8; stratification protects the permutation null but cannot remove gene-specific platform effects, as the advisory review noted. The leave-one-platform-out sensitivity is computed and reported with the result.

10 Reproducibility appendix

Preregistrations (committed and pushed before any run of their arc): `projects/long_covid/prereg/LC_P51_prereg` (commit 42deadfa); `LC_P52_preregistration.md` (commit f81bb92d, amended pre-freeze from the advisory review as recorded in `redirect/lc_p52_judge_consult.md`).

Analysis scripts and results: `scripts/lc_p51_train.py/results/lc_p51_train_result.json` (commit 858e63f0); `scripts/lc_p51_evaluate.py` (locked at 54b852cc)/`results/lc_p51_eval_result.json` (commit 61d3d6a1); `scripts/lc_p52_train.py` (locked at ff9d5463, pre-result bugfix 8420e2e1) / `results/lc_p52_train_result.json` (commit 889be703). Pre-arc nulls: `results/longcovid_p36_result.json`, p37, p38. Annotation sweeps: `scripts/lc_annotate*.py`, `projects/long_covid/results/external/`.

Data locks. Every dataset file carries a SHA-256 digest asserted at script start: GSE226260 counts and matrix (p36), GSE275334 normalized panel (p37), GSE334857 raw archive (p38), GSE251872 raw archive and both platform series matrices (p52). Scripts fail closed on mismatch.

Substance audit. The current reading of `scripts/audit_research_body.py` on this PDF is recorded in the commit message of every manuscript revision; at the scaffold commits it read 1 of 50 substantive pages (5 rendered), then 3 of 50 (7 rendered). The count is updated honestly as chapters land; the gate is claimed only when the audit passes.

11 Statistical methods appendix

11.1 Module scoring

For a candidate module M with measurable gene set G_M , each sample’s module composite is the unweighted mean of per-gene z-scores over G_M . Training-cohort expression is library-size normalized to counts per million, filtered (CPM > 1 in at least 20% of samples), and $\log_2(\text{CPM}+1)$ transformed. In P51, z-scores are

fit across the training samples; in P52, z-scores are fit *within each sequencing platform* and then pooled, the preregistered guard against the cohort's platform split. Evaluation z-scores are fit across all 47 evaluation subjects on $\log_2(\text{normalized}+1)$ values, so no group label enters any scaling.

11.2 Effect sizes and permutation tests

Group contrast is Cohen's D on composites (mean difference); Hedges g adds the small-sample correction. Training selection in P51 used 10,000 label permutations (seed 20260926), one-sided in the observed direction - the disclosed anti-conservatism. P52 used 10,000 *two-sided* permutations with labels permuted *within platform strata* (seed 52), so the null preserves platform membership exactly. Evaluation tests used 10,000 permutations (E1 one-sided in the direction fixed by the untouched training cohort; E2 two-sided). All permutation p -values use the $(b+1)/(B+1)$ convention. Leave-one-patient-out stability requires the sign of D to be unchanged in every fold (37/37 in P51; 27/27 required in P52).

11.3 Eligibility bars, locked a priori

A module is selectable only if at least 20 of its genes are measurable in the training cohort and at least 10 are on the 635-gene evaluation panel. The bars exist because the panel is small and module scores degrade below it; they were fixed in the P51 preregistration before the panel's gene list was consulted for anything but the count.

11.4 Complete module-level results

P51 training (GSE226260, 17 PASC vs 20 recovered; p one-sided observed direction; stable = LOPO sign-stable folds):

module	n genes	D	perm p	stable	selected
L1 _i fna	95	-0.134	0.2548	37/37	no
L2 _i fng	188	-0.164	0.1724	37/37	no
L3 _i nflammatory	153	-0.072	0.2872	37/37	no
L4 _i nfa _n fkf	171	+0.127	0.1721	37/37	no
L5 _i l6 _j akstat3	69	-0.174	0.1072	37/37	no
L6 _c omplement	161	-0.196	0.1111	37/37	no
L7 _o xphos	195	-0.356	0.0432	37/37	yes
L8 _h ypoxia	139	-0.130	0.1024	37/37	no
L9 _i l2 _s tat5	166	-0.106	0.2071	37/37	no

P52 training (GSE251872, 12 PI-ME/CFS vs 15 healthy; p two-sided platform-stratified; no module selected, arc ended at training):

module	n train	n panel	D	perm p	stable	selected
L1 _i fna	95	15	+0.021	0.8917	19/27	no
L2 _i fng	189	52	+0.030	0.7825	26/27	no
L3 _i nflammatory	156	48	-0.028	0.8286	26/27	no
L4 _i nfa _n fkf	179	48	+0.076	0.6851	27/27	no
L5 _i l6 _j akstat3	72	35	-0.023	0.8170	24/27	no
L6 _c omplement	163	30	-0.057	0.5444	26/27	no
L7 _o xphos	199	12	-0.014	0.9217	19/27	no
L8 _h ypoxia	145	17	+0.031	0.6303	27/27	no
L9 _i l2 _s tat5	175	39	+0.074	0.1938	27/27	no

The contrast between the two tables is the lane’s methodological lesson in one view: a nominally suggestive one-sided landscape (P51, where several modules sit near .10–.29) becomes a flat null under the corrected two-sided stratified rule (P52), at smaller n with a platform confound absorbed.

12 Preregistered arc P53: the powered ME/CFS-contrast test

12.1 Cohort selection, made auditable

The arc was approved with GSE227375 (69 participants, exercise-challenge design) as the named training cohort. Pre-analysis reconnaissance changed that, and the trail is recorded here exactly. **GSE227375 was abandoned** for two findings: (i) the series deposits no processed expression data at all - 187 RNA-seq runs exist only as raw SRA reads, outside this lane’s lightweight module design; and (ii) its male arm is platform-confounded (male cases and male controls were sequenced on different platforms), which would have restricted any clean contrast to the female arm (19 vs 19) anyway. **GSE128078 was rejected** as under-powered: 14 cases vs 11 controls at baseline, no material improvement over P52’s 27. **GSE293840 was substituted** (parent-approved before any data touch): 93 ME/CFS vs 75 healthy sedentary controls, $n=168$ - six times P52’s training power - with raw counts deposited. Its cost is analyte: it is plasma *cell-free* RNA, not cellular RNA, and the preregistration locks an interpretation clause for exactly this gap: a failure indicts the signature strategy plus analyte pairing, not the biology of ME/CFS or its separability from Long COVID.

12.2 Training: at $n=168$, everything is signal

Under the frozen protocol (within-batch z-scoring across the cohort’s three sequencing batches, two-sided batch-stratified permutation, 10,000 draws, seed 53, leave-one-out stability required in all 168 folds), *all nine* Hallmark modules separate ME/CFS from healthy cfRNA (two-sided p from .0008 to .011; `results/lc_p53_train_`). The locked rule picks the smallest- p module: L9 (IL2-STAT5 signaling), up in ME/CFS ($D = +0.258$, $p = .0008$, 168/168 stable, 40 of its genes on the evaluation panel; per-batch D positive in all three batches: 0.38, 0.28, 0.08 - note the weakening in the newest batch, reported as locked sensitivity). The contrast with P52’s flat null at $n=27$ is the power lesson the arc was approved to teach.

12.3 Evaluation: transport fails; the contrast “passes” inverted

On the untouched GSE275334 cellular panel (`results/lc_p53_eval_result.json`): **E1 fails** - the cfRNA-trained L9 signature does not transport to cellular ME/CFS at all ($D = -0.124$, one-sided $p = .90$ in the trained direction, $g = -0.453$; if anything the cellular scores run opposite). **E2 passes** - the same signature separates ME/CFS from Long COVID on the panel (two-sided $p = .0126$, $g = -0.929$, clearing the locked $|g| > 0.8$ bar) - *but in the direction inverted from its cfRNA training*. With E1 failed, the preregistered reading holds: this is cellular L9 biology uncoupled from the cfRNA signature, not a transported ME/CFS state discriminator. The win rule (E1 AND E2) is **not met**; the reported comparator leg is $|g_{E2}| - 0.776$ with bootstrap 95% CI $[-0.49, +1.02]$ - straddling zero.

12.4 What three arcs leave behind

The lane’s cumulative, untuned answer: (i) a cellular oxidative-phosphorylation down-signal transports across assay and cell mixture to separate Long COVID from healthy controls (P51 E1) - the one preregistered positive; (ii) nothing preregistered separates Long COVID from ME/CFS as a transported signature (P51 E2, P52 training-stage null, P53 E1); and (iii) the one strong ME/CFS-vs-LC contrast observed (P53 E2, $g = -0.929$) is an inversion artifact of cross-analyte training - a cautionary result for any future liquid-biopsy-to-tissue biomarker pipeline, and a genuine, citable negative.

13 Per-gene annotation synthesis

Table 3 condenses the eight-service primary annotation of the 18 preregistered signature genes (12 L7 oxidative-phosphorylation panel genes from the committed P51 evaluation; 6 published comparator genes). Europe PMC counts are “long COVID”/PASC co-mentions and are honestly sparse across both sets: the lane’s signals are not literature-saturated, which cuts both ways - less prior support, but also less redundancy with published claims.

Table 3: Primary annotation of the 18 signature genes.

gene	set	protein (UniProt, reviewed)	EPMC LC	ChEMBL	Reactome
ACADVL	L7	Very long-chain acyl-CoA dehydrogenase	14	1	2
AIFM1	L7	Apoptosis-inducing factor 1, mitochondr	15	1	-
BAX	L7	Apoptosis regulator BAX	349	8	7
BCL2L1	comparator	Bcl-2-like protein 1	29	10	7
CXCL8	comparator	Interleukin-8	603	1	8
DLD	L7	Dihydrolipoyl dehydrogenase, mitochondr	77	2	13
ECHS1	L7	Enoyl-CoA hydratase, mitochondrial	10	1	7
GOT2	L7	Aspartate aminotransferase, mitochondr	2	1	5
HADHB	L7	Trifunctional enzyme subunit beta, mit	7	1	8
JAK1	comparator	Tyrosine-protein kinase JAK1	351	9	31
LDHA	L7	L-lactate dehydrogenase A chain	98	2	2
MAP3K8	comparator	Mitogen-activated protein kinase kinas	10	1	2
OAT	L7	Ornithine aminotransferase, mitochondr	189	2	1
OSM	comparator	Oncostatin-M	124	6	2
PDHB	L7	Pyruvate dehydrogenase E1 component su	6	1	4
PDK4	L7	[Pyruvate dehydrogenase (acetyl-transf	20	2	3
STAT3	comparator	Signal transducer and activator of tra	954	5	42
VDAC1	L7	Non-selective voltage-gated ion channe	51	1	5

Three patterns stand out. First, the L7 panel is enzymatic and mitochondrial almost to a gene (ACADVL, DLD, ECHS1, GOT2, HADHB, PDHB, PDK4, OAT metabolic enzymes; VDAC1 the outer-membrane channel; AIFM1 and BAX apoptotic mitochondrial effectors; LDHA the glycolytic shunt) - the down-signature is a coherent energy-metabolism module, not a grab-bag, which is exactly what the g:Profiler enrichment confirmed independently. Second, the comparator six are signaling genes (kinases JAK1/MAP3K8, transcription factor STAT3, cytokines CXCL8/OSM, anti-apoptotic BCL2L1) - a regulatory panel where the L7 panel is an effector panel; the two sets are mechanistically complementary views of the same chronic-state axis. Third, druggability concentrates in the comparator set (JAK1 Tclin with an approved inhibitor class; STAT3, MAP3K8, CXCL8, BCL2L1 Tchem) while the L7 enzymes are largely undrugged (ChEMBL target counts near zero) - consistent with the general pattern that metabolic enzymes are poor drug targets and with the lane’s honest position that its positive is a state signal, not a therapeutic hypothesis.

14 Per-dataset provenance deep-dives

14.1 GSE226260: the whole-blood source cohort

This Nature Immunology 2025 series is the lane’s root dataset in two roles: it is the P51 training cohort and it is the source of the published six-gene comparator panel. The series holds 103 libraries; the lane’s sample map (results/longcovid_p36_samples.csv) collapses them to 37 donors - 17 PASC, 20 recovered - one library per person under explicit selection rules recorded in the map itself. Two provenance facts matter for interpretation. First, the six-gene panel was post-selected by the source authors from this very cohort, so its behaviour here is source-context, not independent evidence - the reason the lane never trains on it and

only uses it as a named comparator on the evaluation cohort. Second, the recovered arm is recovered-from-infection, making this the only cohort in the lane whose control arm shares the infection history; the P51 training contrast (PASC vs recovered) is therefore the etiologically cleanest contrast the lane owns, and the oxphos-down selection emerged from the cleanest available comparison.

14.2 GSE275334: the locked evaluation cohort

A 635-gene NanoString PBMC panel over 47 subjects - 15 Long COVID, 18 healthy controls, 14 ME/CFS - and, to the lane's knowledge, the only public cohort carrying Long COVID and ME/CFS arms on one platform. That uniqueness is why it is locked for evaluation in every arc and never trained on: it is the only ground on which the lane's declared differentiator (separation from ME/CFS) can be tested without cross-study confounding between the two disease arms. Its limits are stated wherever it is used: the panel is immune-focused and its metabolic coverage is incidental (the L7 module survives on 12 of its panel genes); the healthy arm is healthy, not recovered-from-infection, so E1-type contrasts here likely overstate what a recovered-control design would give; and the ME/CFS arm's diagnostic purity is unverifiable from GEO metadata. Sample roles are fixed in `results/longcovid_p37_samples.csv` and asserted in every evaluation script.

14.3 GSE334857: the naive-CD4 coverage test

This series contributed the p38 pre-arc test: whether the six-gene list could be measured in naive CD4 T-cells. It could not - CXCL8, the panel's anchor, is below measurement in that compartment - and the test is recorded as a coverage failure rather than a biological result. The negative matters for cohort design: fixed lists discovered in whole blood do not automatically exist in sorted-cell compartments, and any future lane arc using sorted populations must re-derive, not transfer, its features.

14.4 GSE251872: the post-infectious ME/CFS PBMC cohort

The Walitt deep-phenotyping study's PBMC RNA-seq subseries: 12 post-infectious ME/CFS cases and 15 healthy volunteers at baseline, across two sequencing platforms with a 3/7 and 9/8 case-control split. It was chosen for P52 as the only post-infectious, PBMC-matched ME/CFS training cohort, accepting its two weaknesses in writing: tiny n and the platform split, handled by within-platform z-scoring and stratified permutation. Its flat two-sided null (best $p = .194$) is the P52 chapter's result; the cohort's integrity is not in doubt - the power is.

14.5 GSE293840: the powered cfRNA cohort

168 plasma cell-free RNA profiles - 93 ME/CFS, 75 healthy sedentary controls - across three sequencing batches and three collection sites, with raw counts deposited. It became the P53 training cohort after the cohort-selection trail recorded in that chapter, carrying the locked analyte clause. Its rich signal (all nine modules significant) corroborates its source study's claim that cfRNA carries a strong ME/CFS state fingerprint; its transport failure to the cellular panel is the P53 result, not a defect of the cohort. Batch structure is handled by within-batch z-scoring and stratified permutation, and the per-batch effect gradient of the selected module (0.38, 0.28, 0.08) is reported as locked sensitivity rather than smoothed over.

15 Discussion

15.1 The one positive, stated exactly as far as it goes

The oxphos-down signature is a real, preregistered, cross-assay replication: selected in whole blood against recovered controls, transported to PBMC NanoString against healthy controls, $p = .0038$. It joins a growing literature on mitochondrial and energy-metabolism dysfunction in both Long COVID and ME/CFS, and the lane's annotation landscape adds two independent resonances (g:Profiler metabolic coherence of the panel genes; the PRIDE plasma-proteomics oxidative-stress project). What it is not: specific. The same module behaves as a general chronic-state signal - it fails the ME/CFS contrast, and nothing in the lane licenses calling it a Long COVID biomarker. Its honest framing is a robust, transportable chronic-illness metabolic axis in blood, which is a substrate for biomarker research, not a biomarker.

15.2 The separation problem is the field's problem, shown cleanly

Three differently-powered, differently-designed attempts at the ME/CFS contrast failed in three instructive ways: a discovered cellular signature that does not separate (P51), an under-powered training null (P52), and a powered cross-analyte signature that separates only by inversion (P53). The methodological content for the field: (i) healthy-control contrasts will keep "succeeding" while adding nothing clinically - the recovered and ME/CFS control arms are where claims live or die; (ii) cross-analyte transport (cfRNA to cellular) can manufacture significant contrasts with inverted direction, so transport validation must precede contrast testing exactly as the E1-before-E2 rule enforced here; and (iii) the one cohort that can adjudicate separation (GSE275334) has 14 ME/CFS subjects - the field needs a powered two-disease cohort on one platform more than it needs another single-disease signature.

15.3 What the lane would do next, unfrozen

Recorded as direction, not as results: a female-only re-analysis of GSE227375 from raw reads (19 vs 19, clean single-platform) if compute is ever granted; a preregistered test of the P53-inverted cellular L9 contrast as its own hypothesis (it would need a fresh evaluation cohort, since GSE275334 is now spent for that question); and a metabolic-panel evaluation cohort to give the oxphos signal a fair specificity test. None of these is claimed; all are proposed with their preregistrations unwritten.

16 Sample-map construction rules

Every person-level claim in the lane traces to an explicit, committed sample map; this section records the construction rules so the maps are auditable rather than folkloric. For GSE226260, 103 libraries collapse to 37 donors: where a donor contributed multiple libraries (time courses, technical replicates), exactly one library per person enters the training cohort, chosen by the fixed rule recorded in `results/longcovid_p36.samples.csv` (each row carries its selection flag and reason); the training script asserts both the 37-person count and the 17/20 PASC/recovered split and fails closed on any drift. For GSE275334, the 47 subjects map one-to-one to panel columns with disease labels taken from the source's own column annotations into `results/longcovid_p37.samples.csv`; the evaluation scripts assert the 15/18/14 Long COVID/healthy/ME-CFS composition before any test. For GSE251872, group and platform are re-derived from the two platform-specific series matrices' titles at script start (12 PI-ME/CFS, 15 healthy; 3/7 and 9/8 platform splits) and asserted. For GSE293840, subject identity, group, and batch are re-derived from the series matrix titles and characteristics and cross-asserted against the phenotype characteristic (93/75), with the counts-file column set asserted identical to the metadata subject set. The pattern is uniform: no group label is ever copied from a secondary source, and every script re-derives and asserts its cohort before touching expression values.

17 The comparator's provenance, kept separate

The six-gene comparator (JAK1, CXCL8, BCL2L1, OSM, MAP3K8, STAT3) enters the lane from the GSE226260 source publication, where it was selected from that cohort's own data. Three discipline rules follow. First, it is never used for feature discovery: its genes overlap several L-modules by construction of the Hallmark sets (STAT3 in L2/L5; CXCL8 and OSM in L3; MAP3K8 in L4/L5/L9; BCL2L1 in L9), so letting it inform module construction would leak a post-selected list into a frozen candidate space. Second, on the evaluation cohort it is run with the identical machinery as the lane's signature (unweighted z-mean composite, same subjects, same permutation engine), so comparisons are machinery-matched; its evaluation behaviour is a fair like-for-like contrast even though its selection history is not. Third, its own P51 result - strongest on both tests yet below the locked ME/CFS bar ($g = 0.776$) - is reported as context for the bar's meaning, not spun as a lane win: beating a post-selected six-gene list is not the lane's claim, and the P52/P53 preregistrations demoted exactly that comparison from gate to reported statistic on the advisory review's argument.

18 Preregistration discipline as the lane's method

The arcs share one operational skeleton, worth stating once as a method: (i) freeze the question, candidates, eligibility bars, tests, seeds, win rule and stopping rules in a committed preregistration before any run of the arc; (ii) lock analysis code in a committed script before its first execution; (iii) execute once per stage - training, then evaluation only if the training stage's own gate passes; (iv) commit the JSON result with the outcome, whatever it is; (v) disclose implementation bugs with their pre-result status; (vi) carry known statistical weaknesses forward as named clauses in the next preregistration rather than silently fixing them. The lane's ledger shows the skeleton doing its work: P51's one-sided-selection flaw was disclosed in its result commit and became P52's mandated two-sided correction; P52's advisory review demoted a stacked win gate before it could produce a spurious win; P53's analyte clause converted what could have been an inverted-direction "win" into an honest strategy-plus-analyte negative. None of these protections cost a real positive: the oxphos E1 result survives all of them.

19 Inside the E1 composite (post-hoc, descriptive)

The following decomposition of the committed P51 evaluation is post-hoc and descriptive: no selection, threshold or decision depends on it, and it is not a gated test. It answers the honest reviewer's question - is the composite carried by one gene? - with per-gene Hedges g on the evaluation cohort (negative = lower in the first-named group).

L7 panel gene	g LC vs healthy	g LC vs ME/CFS
ECHS1	-1.39	+0.05
AIFM1	-0.74	+0.27
BAX	-0.66	-0.15
VDAC1	-0.64	+0.22
OAT	-0.40	+0.23
PDK4	-0.35	-0.70
HADHB	-0.27	-0.39
ACADVL	-0.25	-0.64
LDHA	-0.18	+0.02
GOT2	-0.01	+0.17
PDHB	+0.13	-0.07
DLD	+0.18	-0.16

Two readings. First, the healthy contrast is broad-based, not single-gene: ten of twelve genes are negative

(down in Long COVID), led by ECHS1 ($g = -1.39$), AIFM1 (-0.74), BAX (-0.66) and VDAC1 (-0.64), with only DLD and PDHB mildly positive - the composite's E1 pass rests on a coherent module-level down-shift, which is what a pathway-level signal should look like. Second, against ME/CFS the structure dissolves: signs mix (ACADVL and PDK4 down, VDAC1/AIFM1/BAX up, the rest near zero), which is exactly what the E2 failure says at module level. The two columns together are the lane's whole story in miniature: coherent against health, incoherent against the disease neighbour.

20 The pre-arc nulls in detail

20.1 p36: the six genes in their own source cohort

In GSE226260 itself (17 PASC vs 20 recovered, 12,828 measurable symbols after mapping; `results/longcovid_p36_re`) the published six-gene list shows 5 of 6 expected up-signs against an empirical null mean of 1.70 (10,000 random six-gene sets), $p = .0081$ - nominally strong, but the strict registered rule requires all six, and no individual gene survives a Bonferroni bar. A post-selected list showing a one-gene-miss in its own discovery cohort is the textbook signature of selection optimism; the lane records it as a weak source-context signal and nothing more.

20.2 p37: the same list cross-assay

On GSE275334 PBMC NanoString (15 Long COVID vs 18 healthy; `results/longcovid_p37_result.json`), per-gene Welch tests give: JAK1 $g = 0.90$ ($p = .010$), CXCL8 $g = 0.44$ ($p = .20$), BCL2L1 $g = -0.11$ ($p = .75$, wrong direction), OSM $g = 0.53$ ($p = .14$), MAP3K8 $g = 1.19$ ($p = .0011$, the one gene passing Bonferroni), STAT3 $g = 0.28$ ($p = .40$). The registered cross-assay support flag is FALSE: five of six up but only one gene multiplicity-safe, and the committed caveat states why even a clean pass would have been limited - source genes post-selected from whole blood, and PBMC-NanoString LC-versus-healthy is not whole-blood PASC-versus-recovered. This is the result that made the lane adopt module-level preregistered arcs.

20.3 p38: the coverage failure

On GSE334857 naive CD4 T-cells (10 people; 25 records; `results/longcovid_p38_result.json`), five of the six genes are measurable in all or nearly all samples, but CXCL8 - the panel's anchor cytokine - is above 1 CPM in zero of ten. A six-gene rule with an unmeasurable anchor is untestable by construction; recorded as a coverage failure, and the basis for the lane's rule that sorted-compartment questions need re-derived features.

21 Conclusion

The lane set out to test whether Long COVID blood expression could carry a preregistered, transportable signal, and whether that signal could do the one thing that would make it clinically interesting: separate Long COVID from ME/CFS. It answers the first question with a qualified yes - an oxidative-phosphorylation down-module, selected under a frozen protocol and transported cross-assay against healthy controls ($p = .0038$, broad-based across its panel genes) - and the second with three disciplined negatives that are themselves the lane's most durable output: a discovered cellular signature that does not separate (P51 E2), an under-powered training null honestly banked (P52), and a powered cross-analyte signature whose significant separation arrives inverted, exposing a transport artifact class that any liquid-biopsy biomarker program must design against (P53). The gates stand exactly where the evidence puts them: records and services met at their stated units; the paper below its floor and labeled so; the benchmark endpoint open. The field's missing instrument, on this evidence, is not another single-disease signature but a powered two-disease

cohort on one platform - until then, the honest claim is a chronic-state metabolic axis in blood, robustly measurable, not yet a Long COVID biomarker.

22 Hub-level annotation narrative

The tertiary sweeps annotate the six comparator hubs across two dozen services; the scientifically legible content, beyond the ledger counts, is as follows (all evidence in `projects/long_covid/results/external/tertiary`).

Druggability is real but concentrated. DGIdb returns 82 drug-gene interactions for JAK1 and 80 for CXCL8, against 43 for STAT3, 14 for BCL2L1 and 3 each for OSM and MAP3K8; Pharos development levels agree (JAK1 Tclin, CXCL8/BCL2L1/MAP3K8/STAT3 Tchem, OSM Tbio), and the approved JAK inhibitor class (tofacitinib, OpenFDA label retrieved) is the only immediately clinically-mapped entry point in the whole 18-gene signature universe. A biomarker lane this is not; a repurposing hypothesis generator it could become, but only after a specificity result that does not yet exist.

Genetic evidence is thin where it should be thin. GWAS Catalog mapped-gene association counts (JAK1 139, STAT3 94, CXCL8 78, BCL2L1 74, OSM 50, MAP3K8 20; bogus-gene control 0) reflect these genes' general immune-disease pleiotropy, not Long COVID genetics - and the PGS Catalog holds zero Long COVID traits, an honest empty that says the genetic arm of the field has not yet produced a polygenic score to test against.

Variation and regulation context. ClinVar counts (JAK1 922, STAT3 996, MAP3K8 311, OSM 85, BCL2L1 43, CXCL8 42) track gene size and clinical salience rather than disease relevance; Signor signed-network edges (STAT3 167, JAK1 76, BCL2L1 44) confirm the comparator set is a signaling core; validated miRNA regulator counts from miRTarBase (MAP3K8 39, CXCL8 35, BCL2L1 34, JAK1 32, STAT3 25, OSM 7) are recorded as regulatory context for any future post-transcriptional arc. None of this is evidence for or against the lane's preregistered results; it is the annotated ground on which a follow-up program would build, deposited so the next lane starts from evidence rather than search.

23 Related work, as the lane's datasets see it

The lane's literature position is defined by its four cohorts' source studies rather than by a systematic review, and it is stated that way honestly. The GSE226260 source study (Nature Immunology 2025) established whole-blood transcriptomic differences between PASC and recovered donors and contributes the six-gene panel this lane uses strictly as a named comparator - the lane's P36 result shows that panel is weaker in its own source cohort than its selection history suggests. The GSE275334 source study (JCI Insight) built the 635-gene PBMC panel this lane treats as locked ground, and is, to the lane's knowledge, the only public study carrying Long COVID and ME/CFS arms on one platform - the design the field needs at ten times the size. The Walitt deep-phenotyping study (GSE251872 and parent superseries) characterised post-infectious ME/CFS with rigorous clinical criteria and modest transcriptomic findings - consistent with the lane's P52 flat null at that n . The cfRNA study behind GSE293840 reported diagnostic-grade machine-learning performance from plasma cell-free RNA - consistent with the lane's P53 finding that all nine modules separate ME/CFS cfRNA from healthy, and equally consistent with its transport failure, since a classifier trained and validated within one analyte never has to cross compartments. Where the lane adds to this landscape: module-level preregistered selection with patient-level stability, cross-assay transport as an explicit frozen test, the ME/CFS contrast treated as the load-bearing control rather than an afterthought, and every negative kept on the record. Where it deliberately does not compete: single-cohort classifier performance, where the cfRNA study's own numbers already exceed anything this lane's panel evaluation could support, and where overfitting risk is highest.